

Testing the Streaming Instability with the KBO size distribution

Weston Hall

Jake Simon

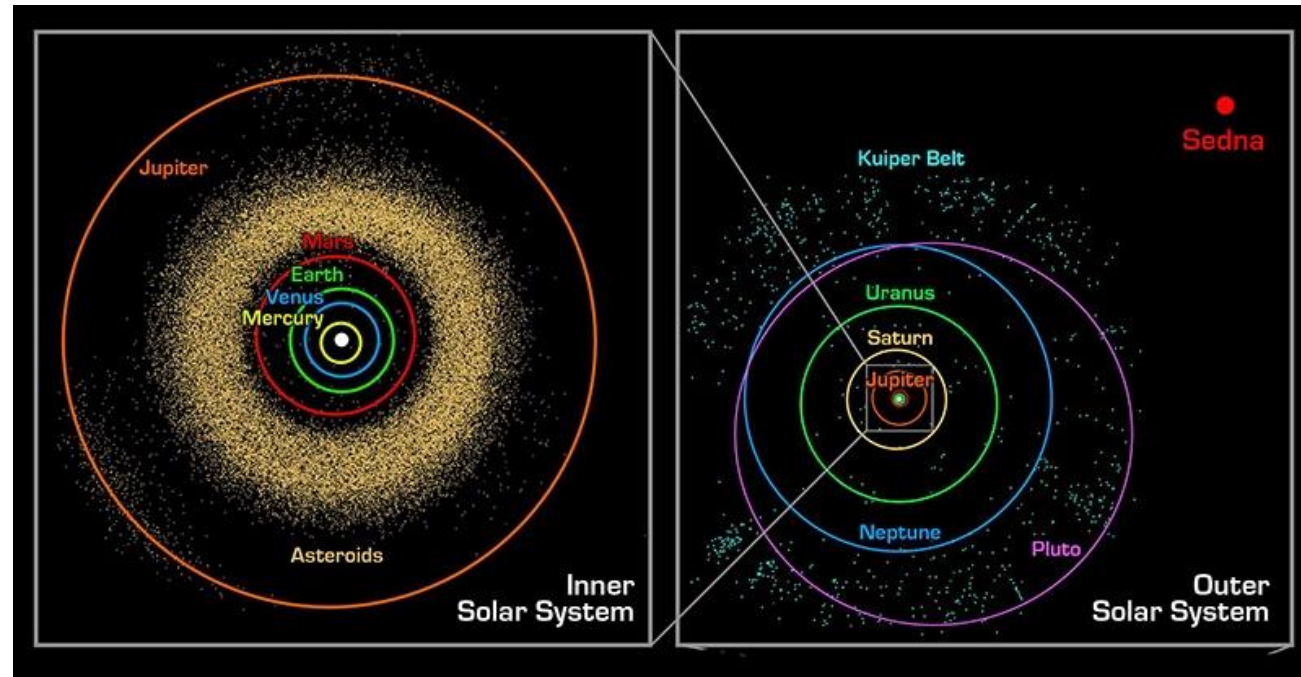
3rd PFITS+ Meeting

06/08/2026

Planetesimal populations

Main Belt

- Asteroid collisions common
- Asteroids formed from fragments common



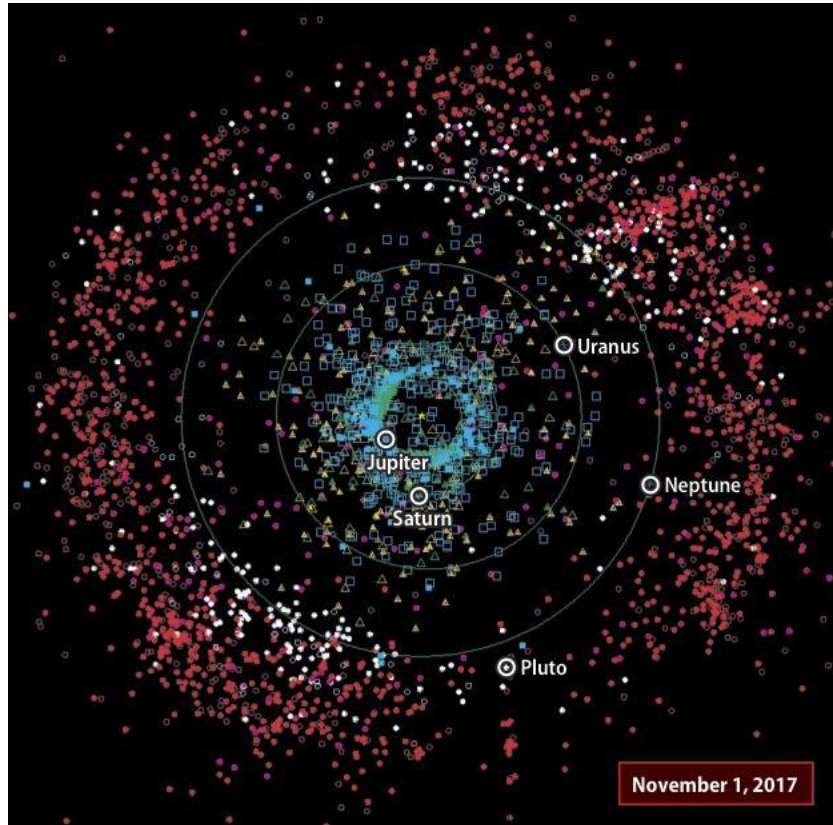
NASA

Kuiper Belt

- Fewer asteroid collisions
- Multiple populations

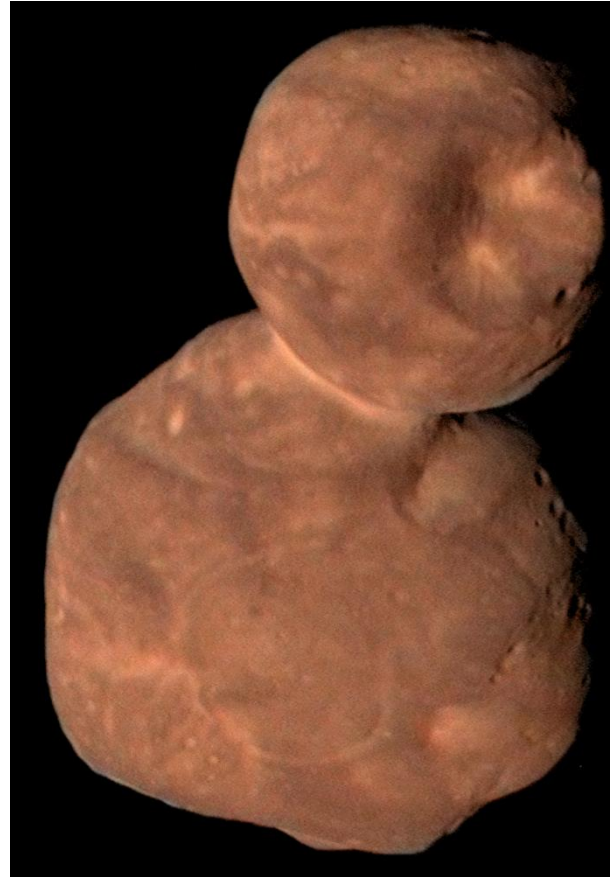
Others include: Detached objects, Trojan asteroids

Of interest: Cold Classical Kuiper Belt (CCKB)



CfA

Subset of Kuiper Belt



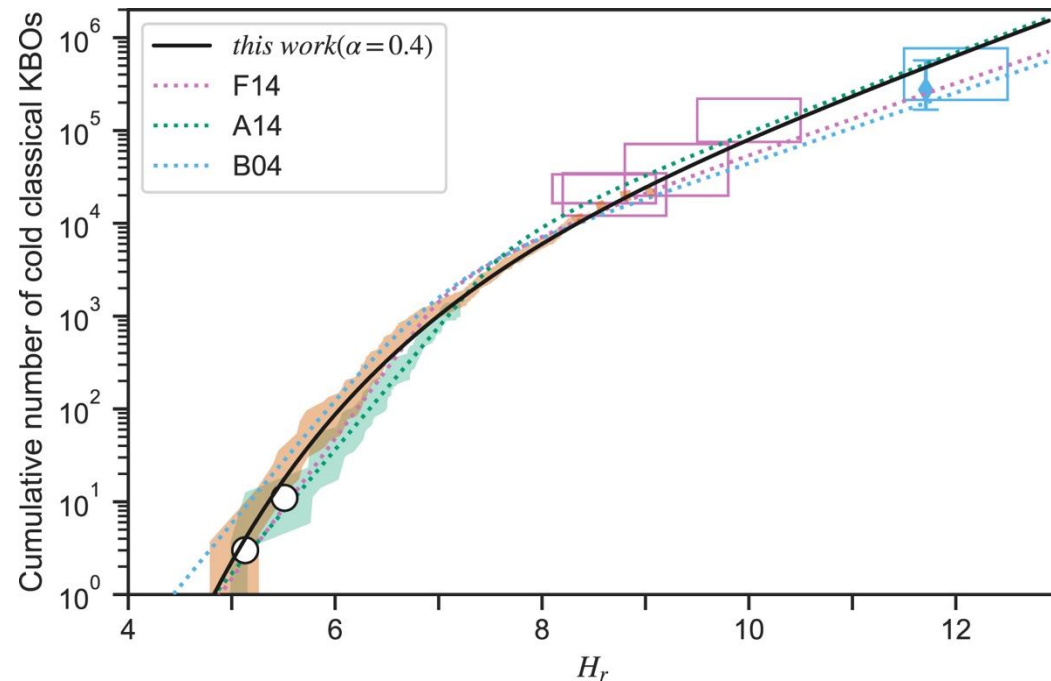
486958 Arrokoth

Attributes:

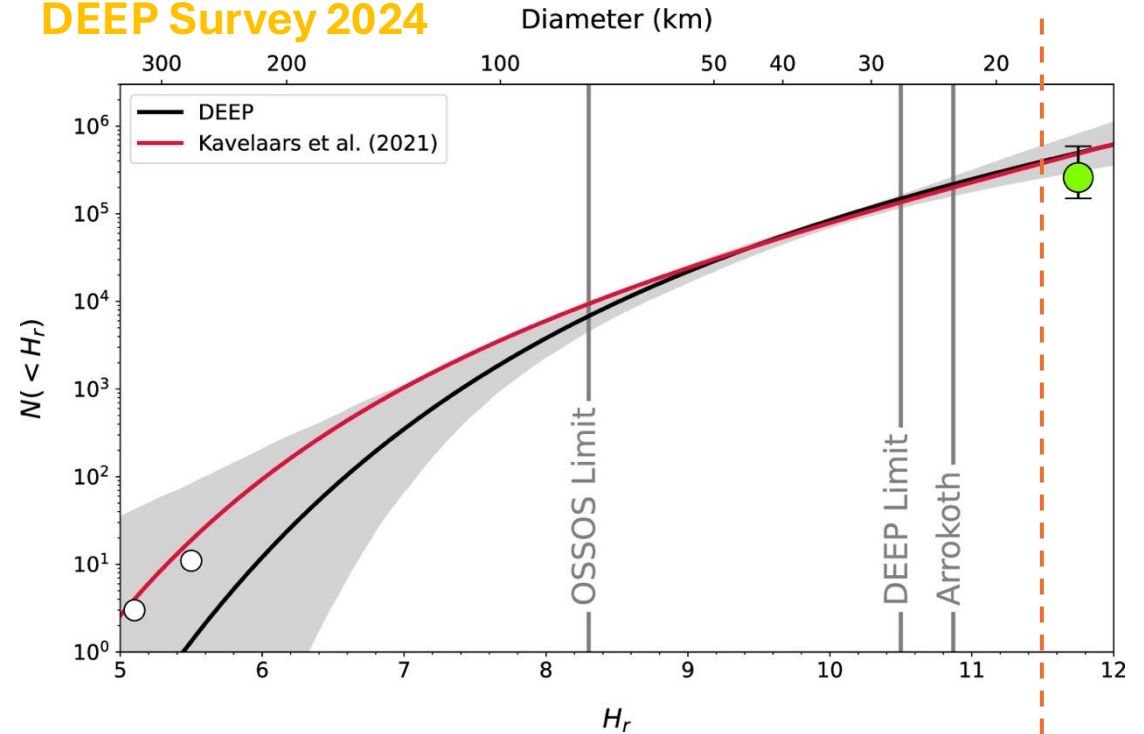
- Dynamically cold
 - Low inclination and eccentricity
- Low collisions
 - Combined with dynamics, suggests *in situ* formation
- High binary fraction
 - Some contact binaries
- Distinct composition and sizes
- **Primordial**

The CCKB has been observed to have an exponential size taper in magnitude distribution

OSSOS Survey 2021



DEEP Survey 2024



Note: **H magnitude** on x-axis

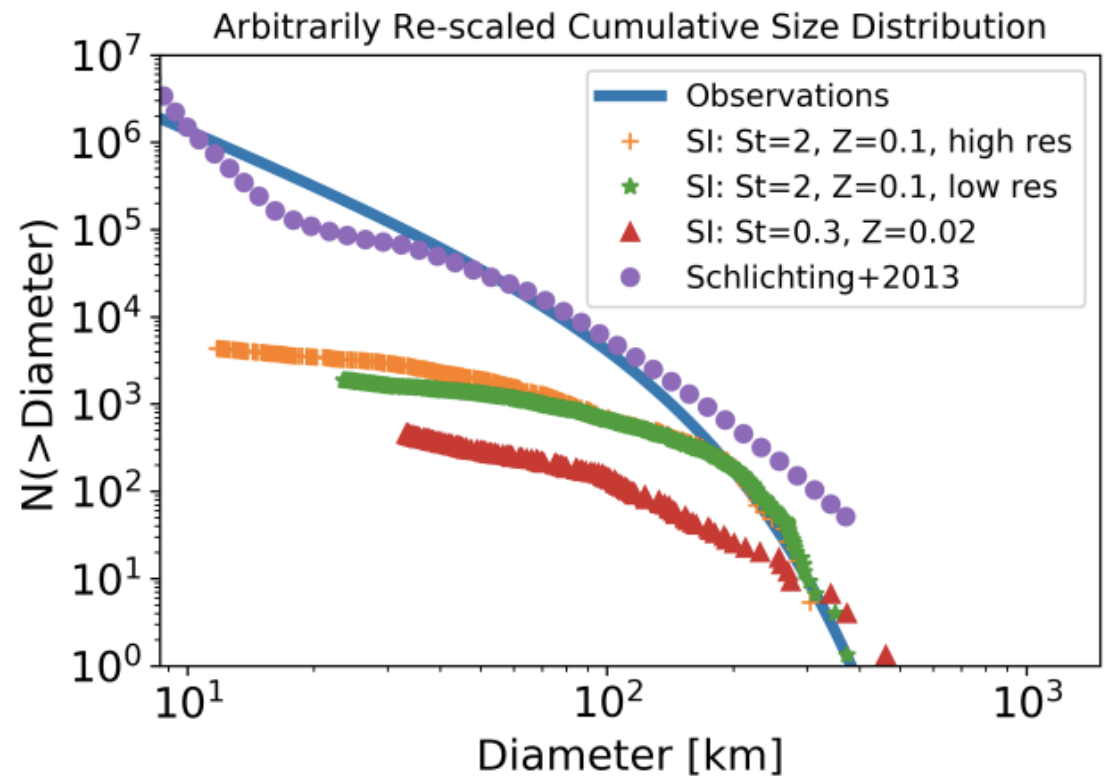
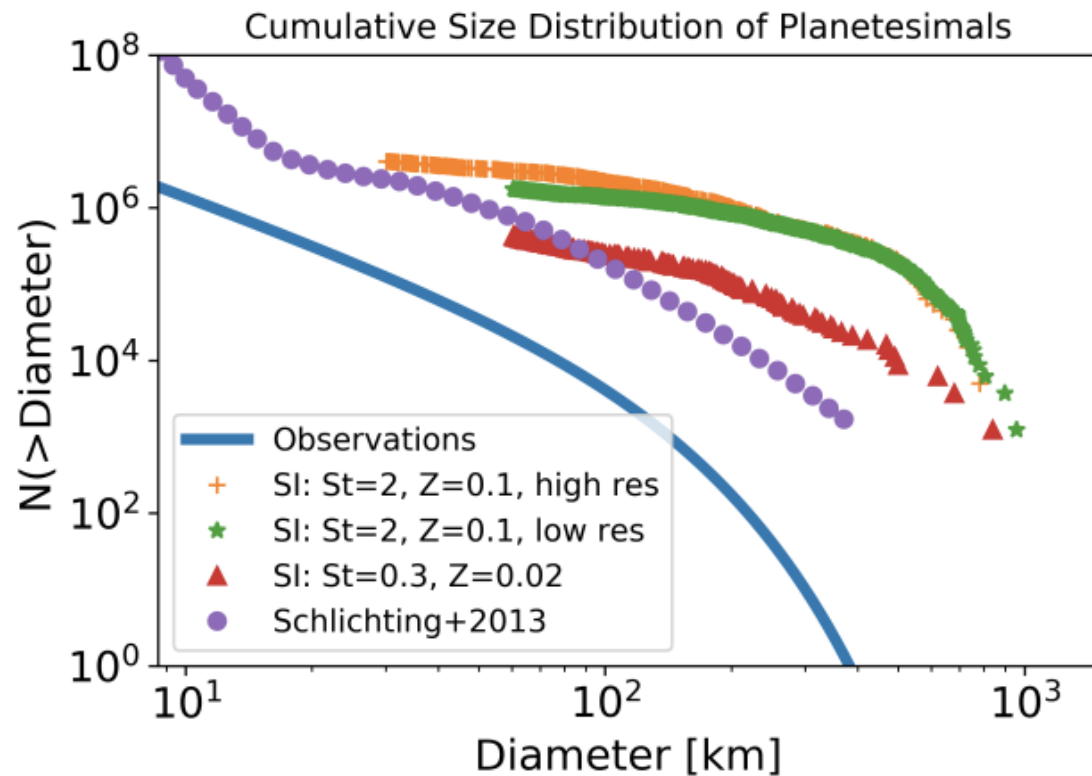
$$N(< H_r) = 10^{\frac{3}{5}\alpha_{SI}(H_r - H_o)} \exp\left[10^{-\frac{3}{5}\beta_{SI}(H_r - H_B)}\right]$$

Vera Rubin (LSST)
Limit
2026-2027

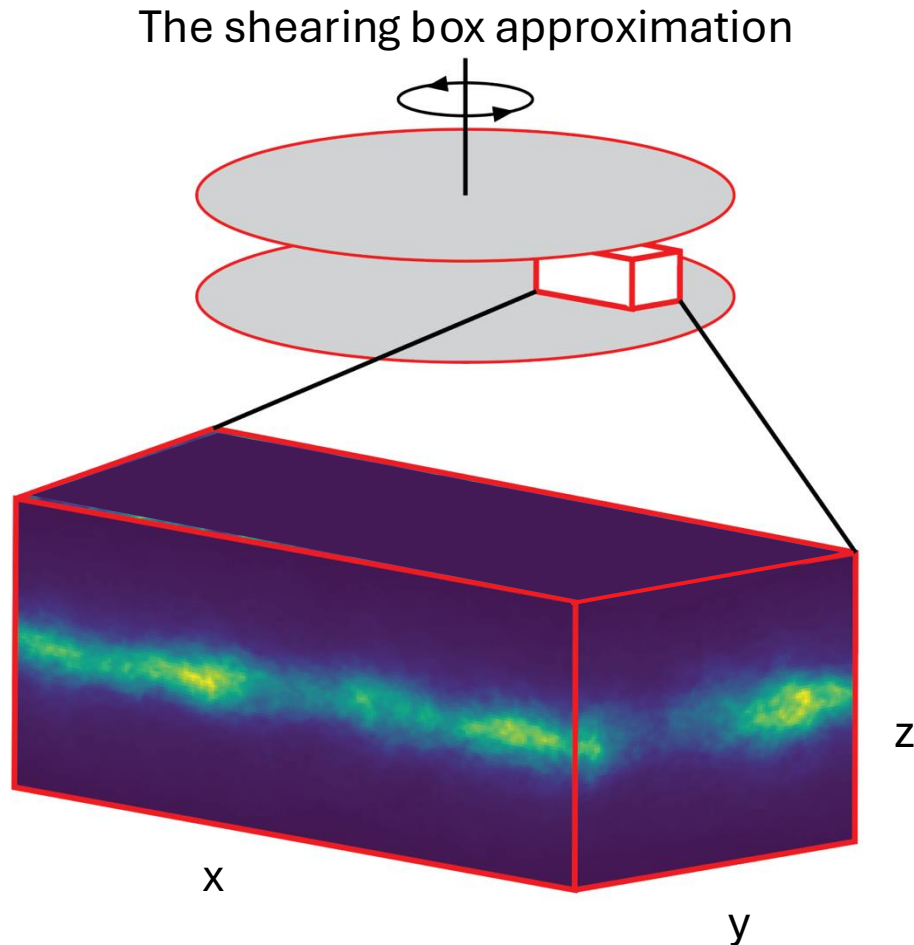
Could the Streaming Instability create the primordial planetesimal population of our Solar System?

Specifically: does the size distribution from theory match the size distribution from observation?

Previous SI simulations (fewer parameters explored, less rigorous clump tracking) appear to not match the CCKB. Is this true with better treatments?

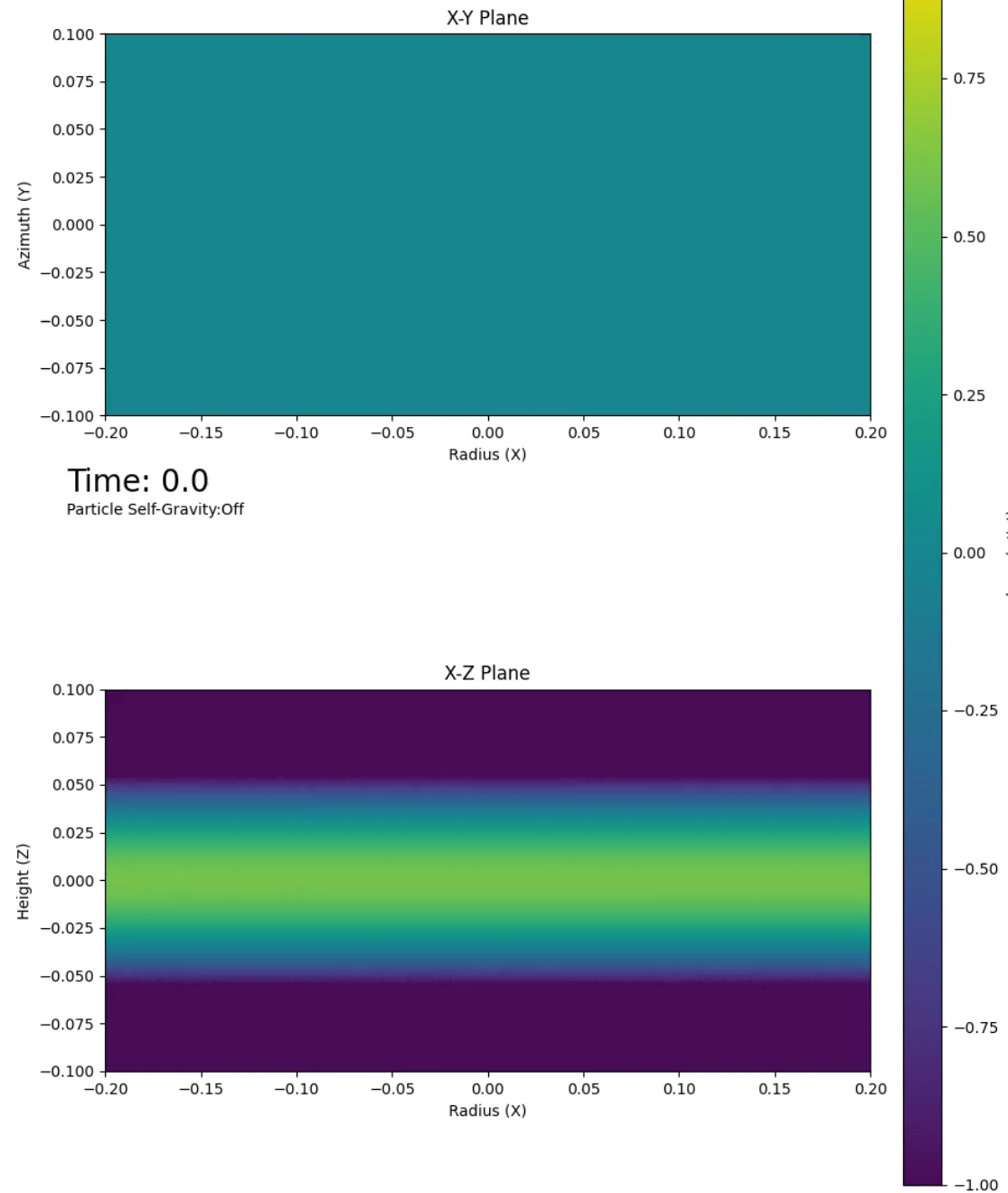


Hydrodynamic simulation with *Athena*



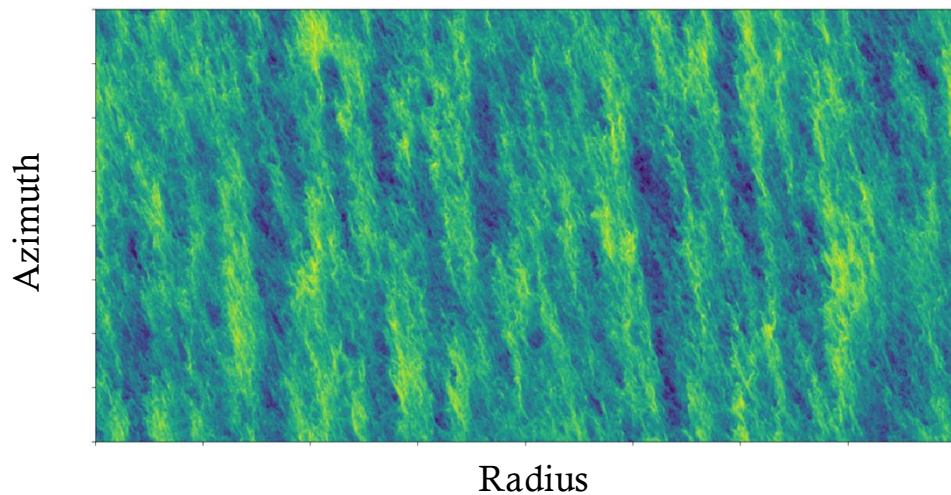
- Gas dynamics equations solved numerically
- Includes a solid component represented by Lagrangian particles
- Simulate a “shearing box” – small corotating patch of the disk
- Particle self-gravity is “turned on” at some time in the simulation
- Different simulations are characterized by different physical parameters

$Z = 0.02$
 $\tau = 0.1$
 $\Pi = 0.05$
 $\tilde{G} = 0.1$

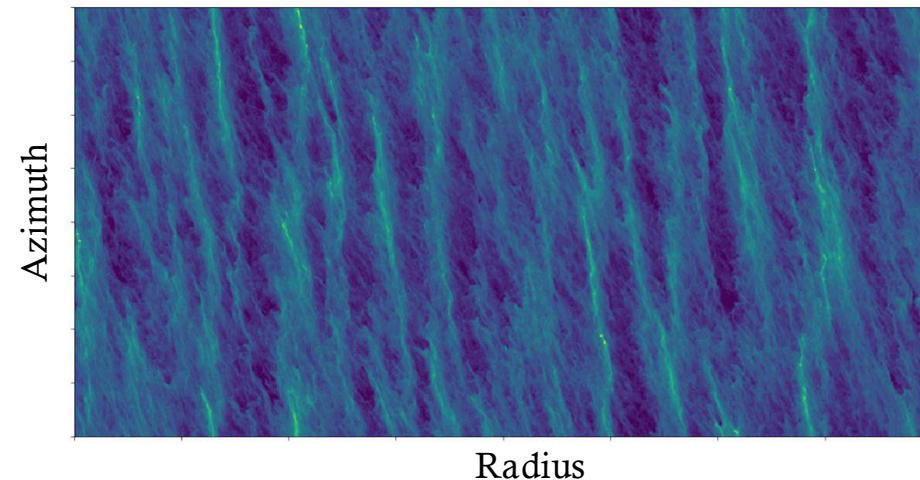


Particle self-gravity
activates at Time: 30

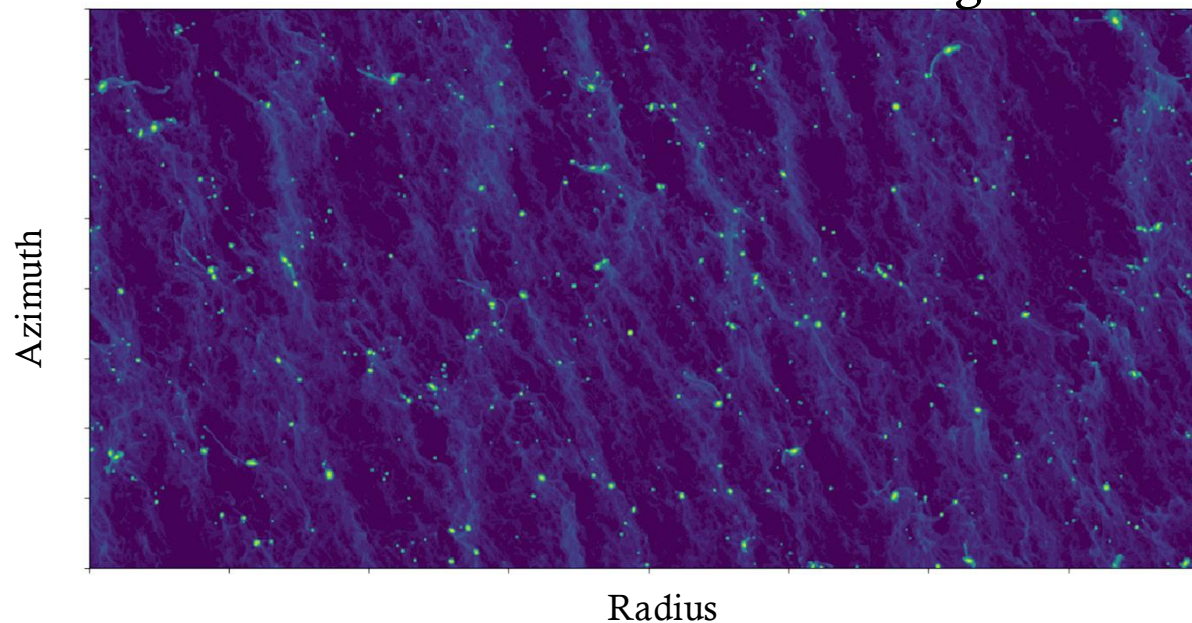
Before SG activation



After SG activation



After simulation has run a long time



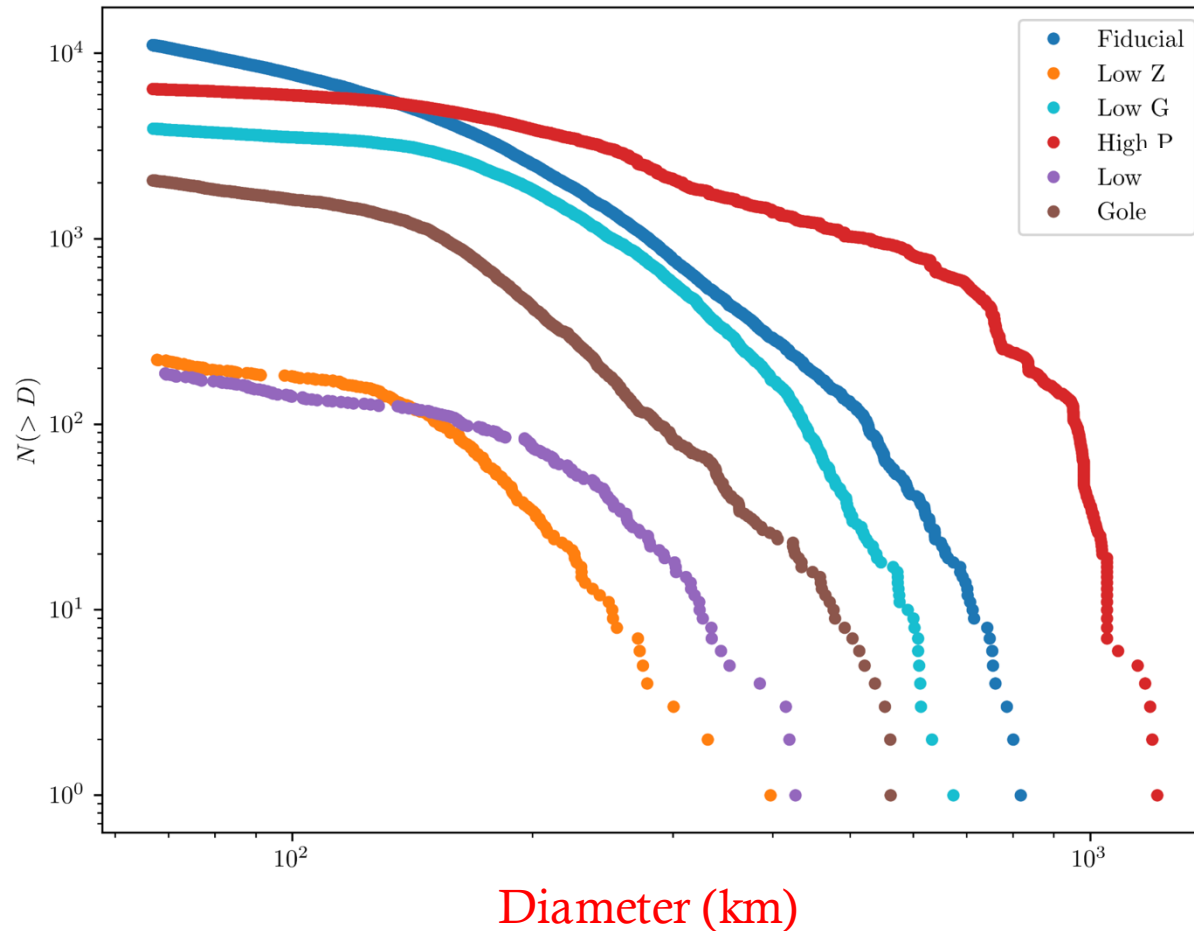
Areas that pass Roche density can be identified as clumps/planetesimals

Bound planetesimal/clump **masses** are calculated using the solid particles they contain

Note that these images are “snapshots” of the simulation

Key improvement: Use of a clump tracking algorithm to select only new/primordial planetesimals

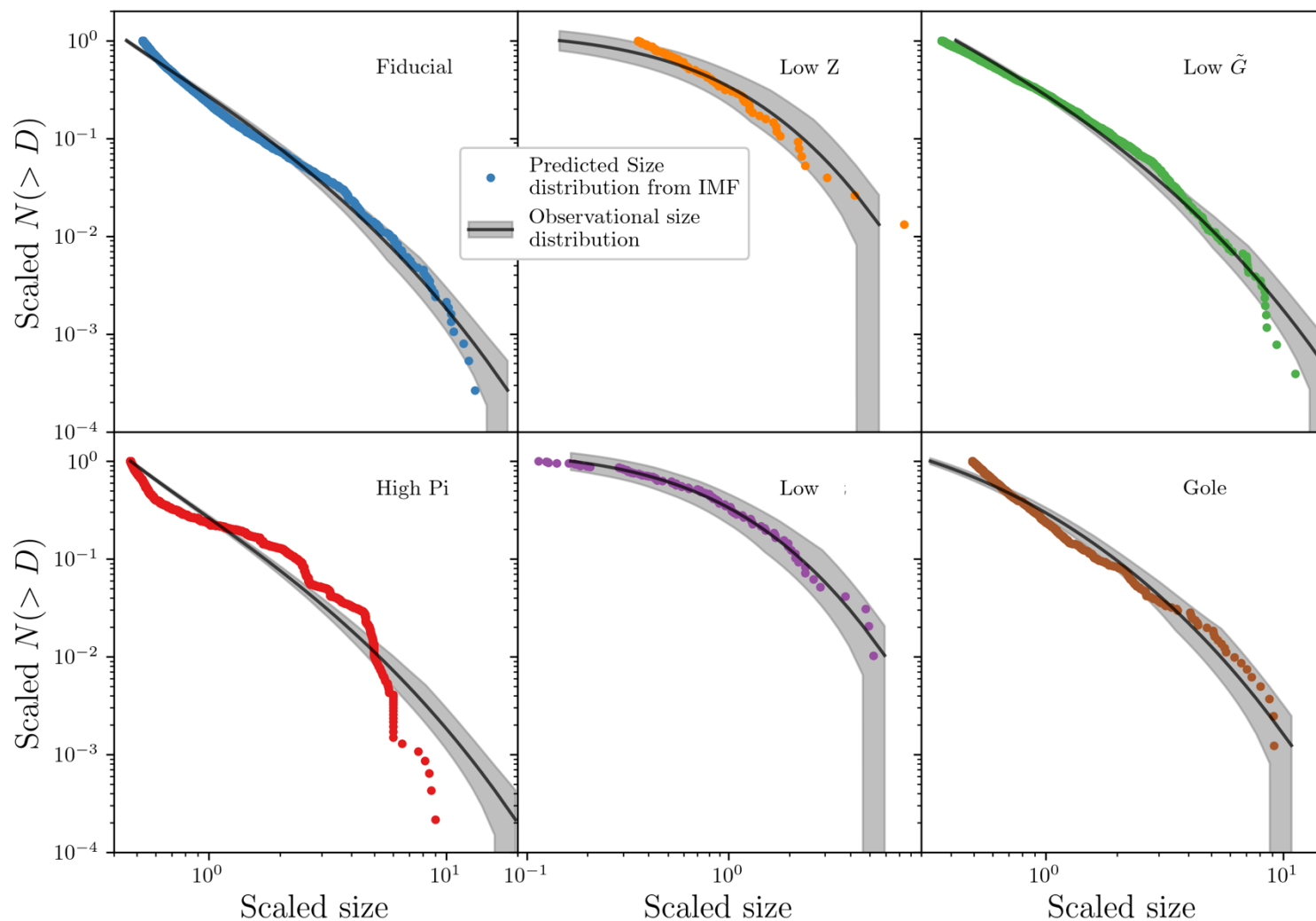
Table of runs



Name	$N_x \times N_y \times N_z$	τ	Π	Z	$\tilde{G}$
Fiducial*	$1024 \times 512 \times 512$	0.1	0.05	0.05	0.1
Low Z	$1024 \times 512 \times 512$	0.1	0.05	0.02	0.1
Low $\tilde{G}$	$1024 \times 512 \times 512$	0.1	0.05	0.05	0.05
High Pi	$1024 \times 512 \times 512$	0.1	0.1	0.05	0.1
Low	$1024 \times 512 \times 512$	0.01	0.05	0.05	0.1
Gole	$1024 \times 512 \times 512$	0.3	0.05	0.02	0.1

Note: **Masses** were identified from *Athena* sims, observations are done in **Magnitude** (brightness), so we will perform comparison in **Diameter (size)** space, using assumptions about the shape, composition, and formation environment.

1st Comparison: Sampling observations



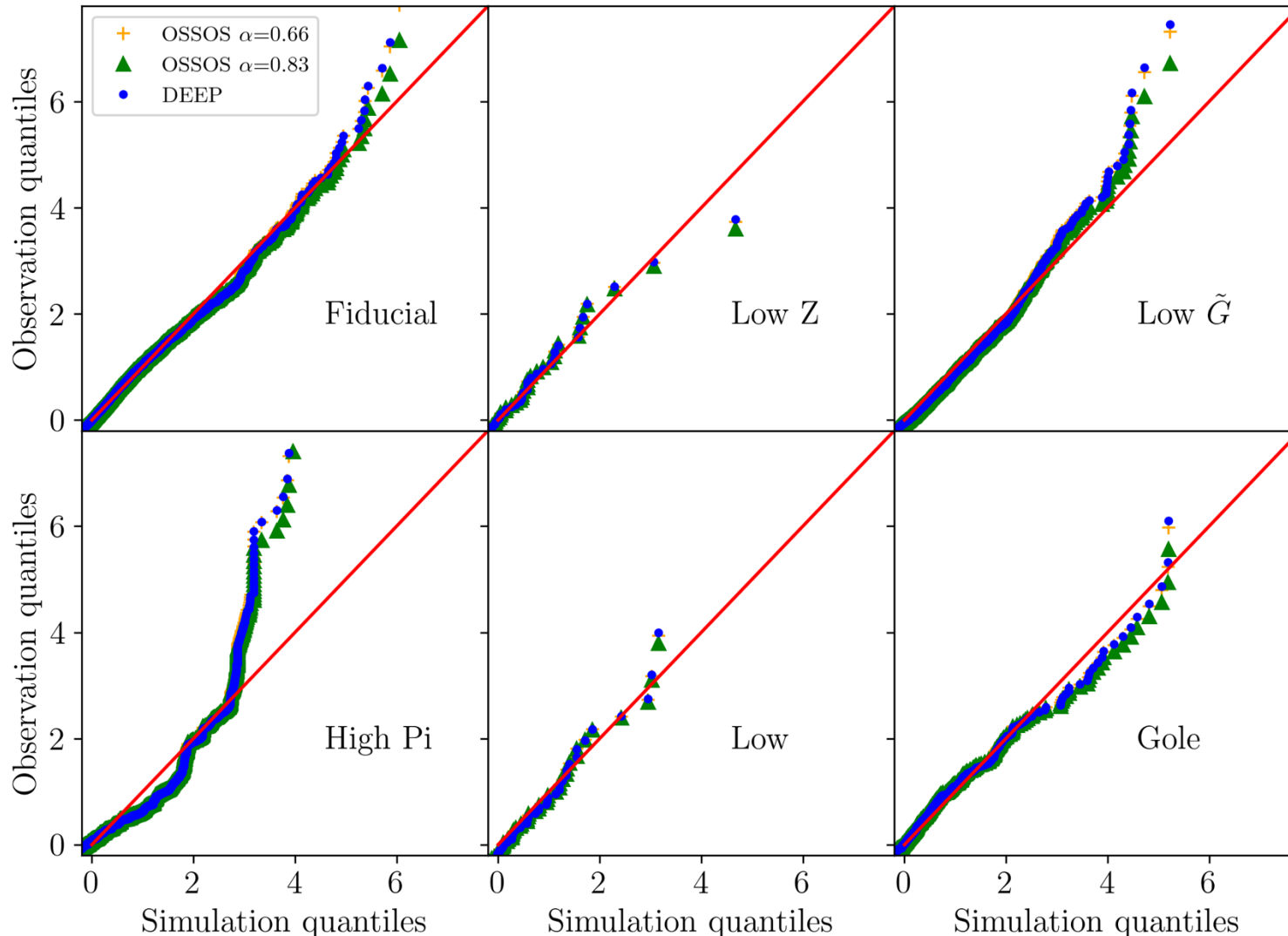
Recall: Observation is a full distribution, simulation is sample

Sample repeatedly from observation, scale sizes and numbers to reduce uncertainties.

	Objects	OSSOS 1 N/A	OSSOS 2 N/A	DEEP N/A
Fiducial	3762	~ 0	~ 0	~ 0
Low Z	76	0.002	0.002	0.002
Low $\tilde{G}$	2567	~ 0	0.265	0.047
High Pi	4644	~ 0	~ 0	~ 0
Low	97	0.899	0.563	0.899
Gole	808	~ 0	~ 0	~ 0

Key takeaway: Visual similarity between samples, statistical similarity. This motivates a shape-based comparison

2nd Comparison: Distribution shape

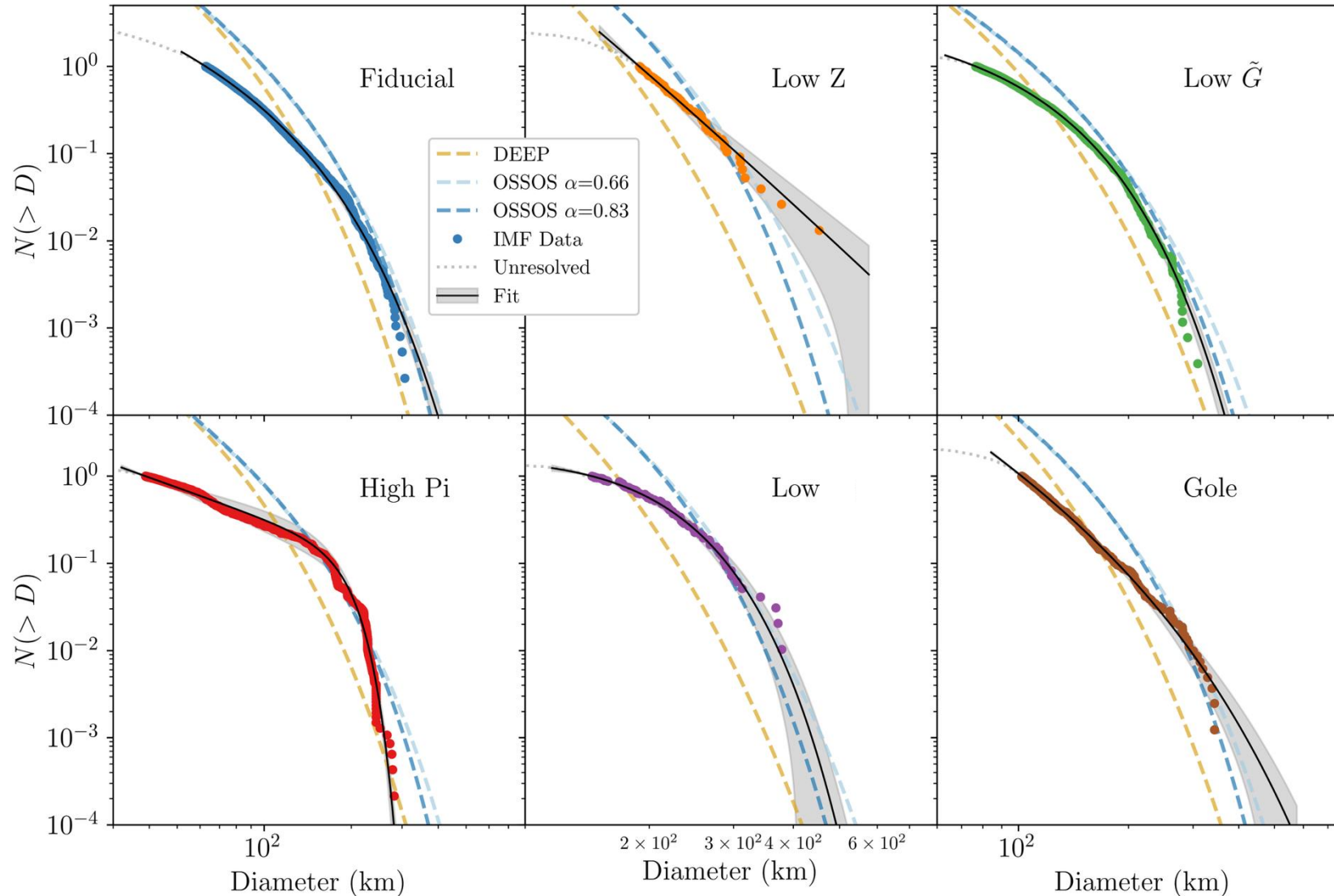


Instead of sampling, shapes can be compared directly by transforming the distribution.

Observation has distinctive exponential taper – look for that agreement

Key takeaway: The shape of the distribution possibly matches, meaning the simulation samples don't appear biased. Motivates parametrizing the shape.

3rd comparison: Parametrized form



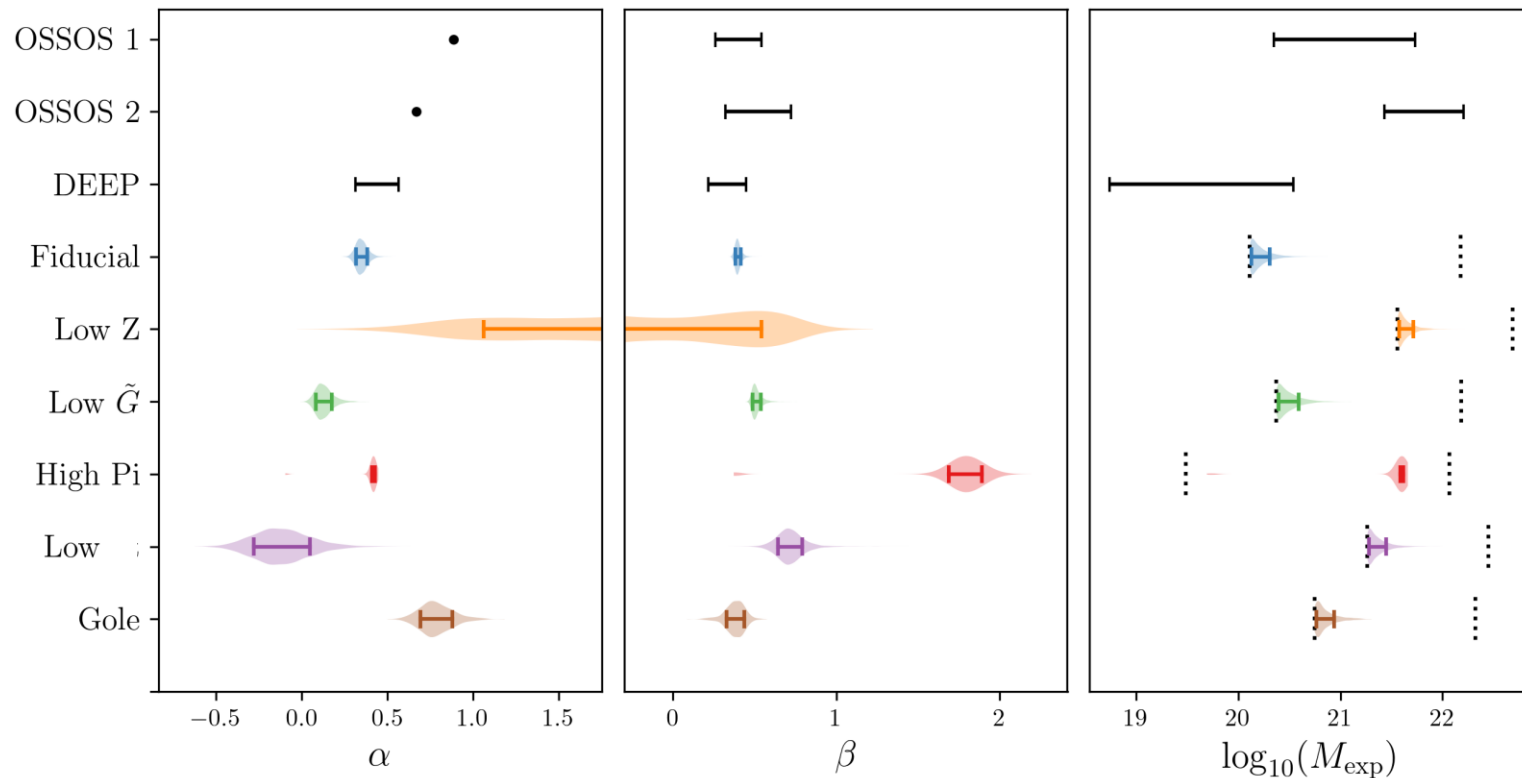
Full uncertainty consideration. Fit to IMF as if it were full distribution.

VTPL:

$$P(> N, \theta) = cM^{-\alpha} \exp \left[- \left(\frac{M}{M_{\text{exp}}} \right)^{-\beta} \right]$$

α : power law exponent, describes distribution at small sizes
 $M_{\text{exp}}/D_{\text{exp}}$: "exponential" size, location where tapering starts
 β : Tapering strength, describes distribution at large sizes

$$P(> N, \boldsymbol{\theta}) = cM^{-\alpha} \exp \left[- \left(\frac{M}{M_{\text{exp}}} \right)^{-\beta} \right]$$

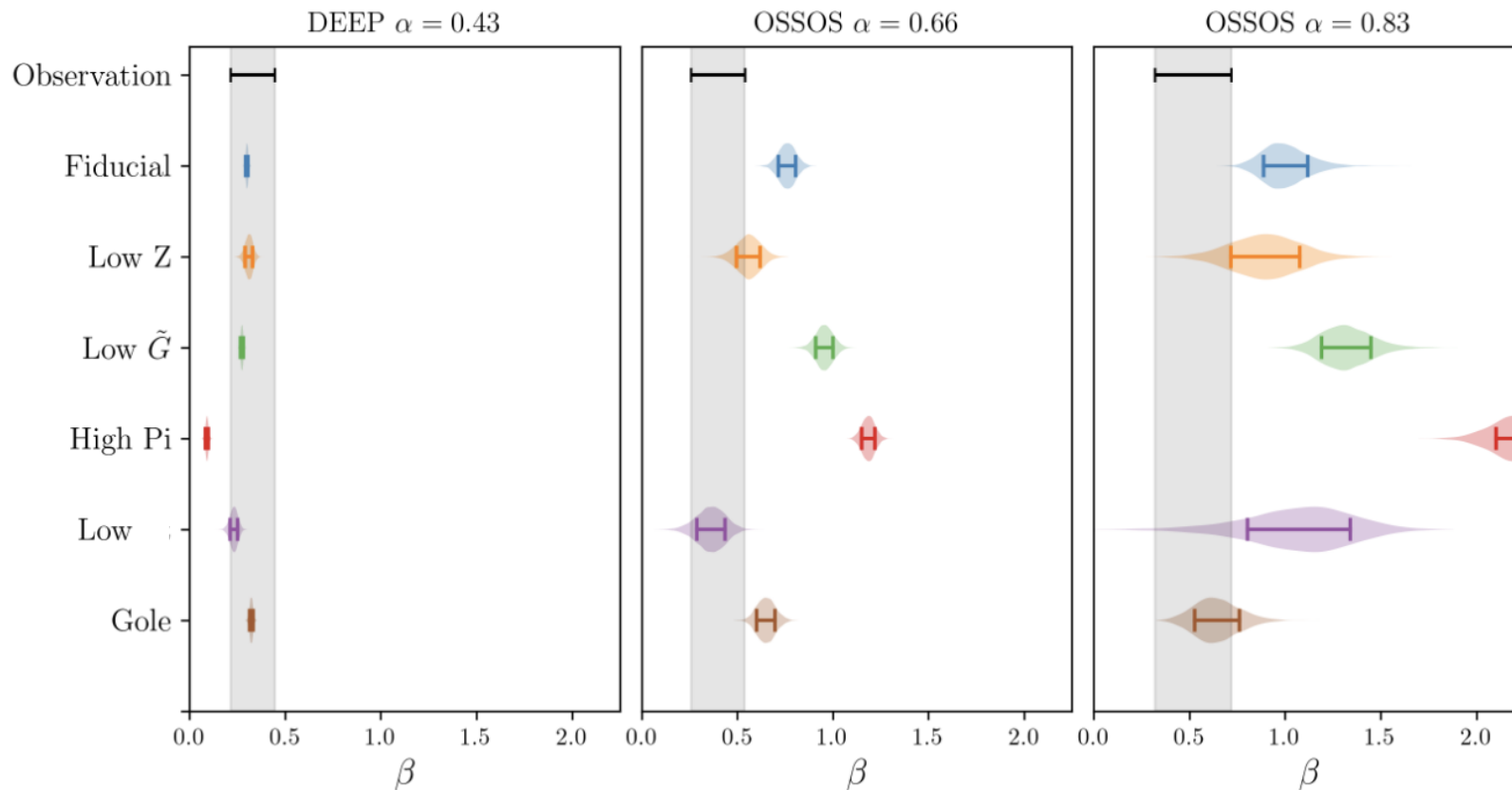


VTPL fitting is poor at selecting a characteristic mass/size, probably because mass scale of the IMF is too low.

This leads to poor uncertainties on other parameters.

Key takeaway: Small edge of IMF is fitted poorly with parameters. Maybe lower α is preferred.

$$P(> N, \boldsymbol{\theta}) = cM^{-\alpha} \exp \left[- \left(\frac{M}{M_{\text{exp}}} \right)^{-\beta} \right]$$



Same parameter fitted x3 ways

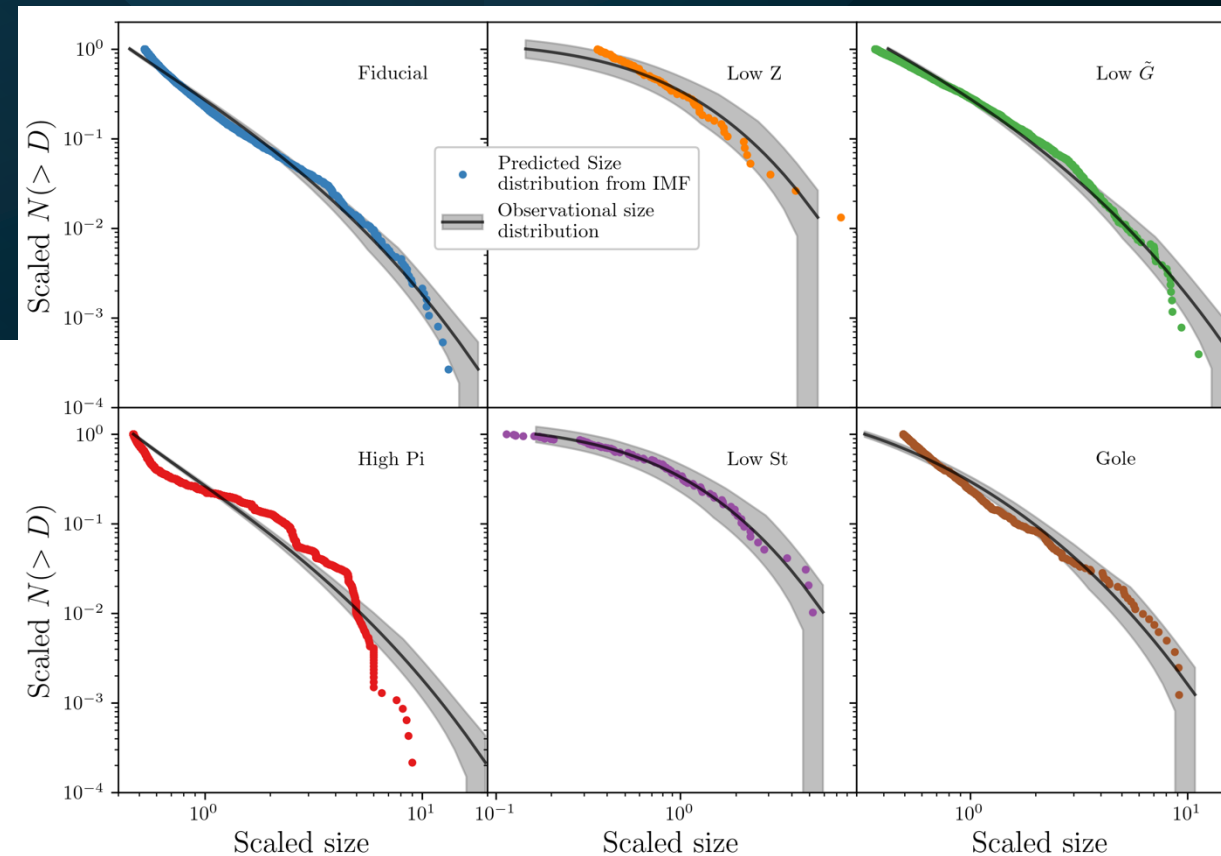
Since fitting is poor at describing the small edge of the distribution, those parameters can be fixed to observational values, and the tapering parameter is fitted alone.

Key takeaway: Simulations are reproducing β , and the constraints are smaller

Conclusion: Could the Streaming Instability create the primordial planetesimal population's size distribution?

Yes*

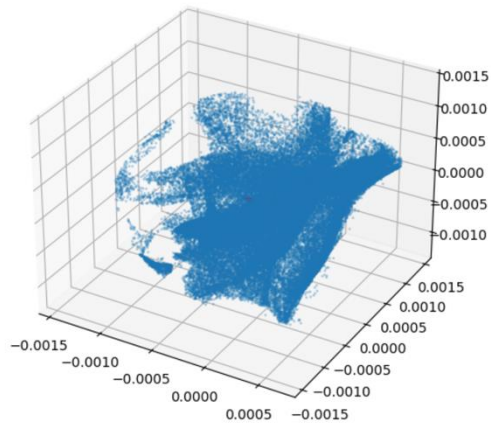
*-Within assumptions and limitations



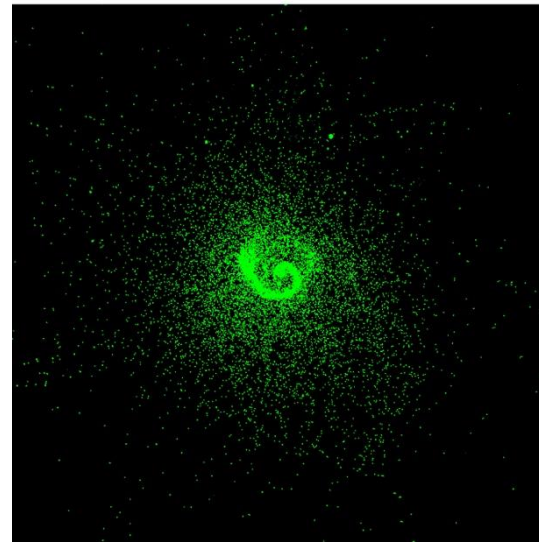
Buffer slide

Follow the gravitational collapse of SI pebble clouds

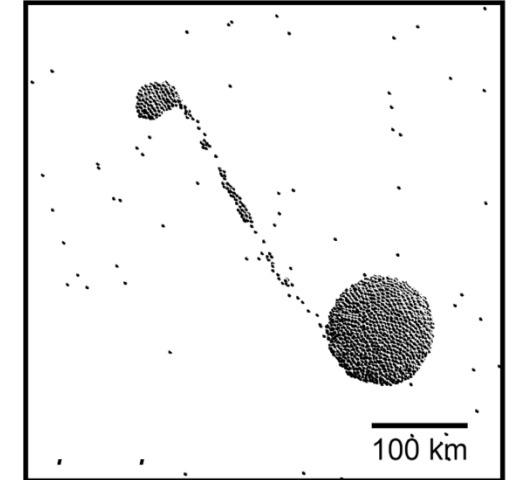
Currently working towards the most realistic gravitational collapse system yet. Using  with custom physics modules.



Gravity + Gas + Collisional energy loss



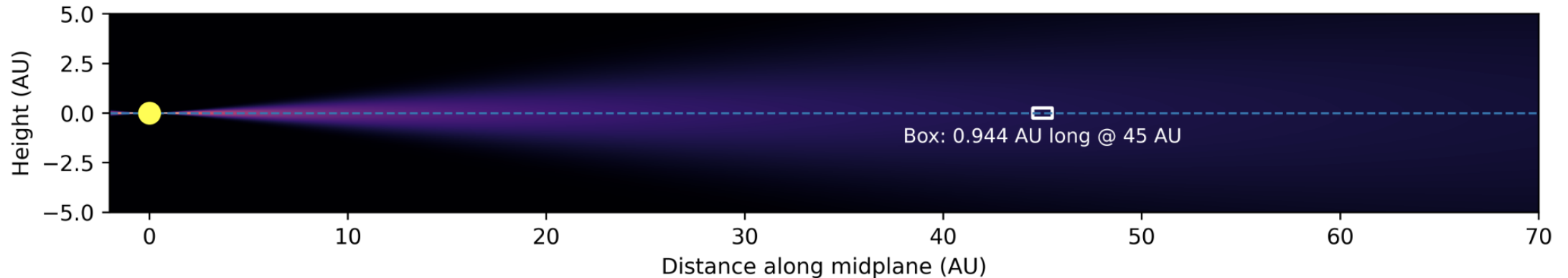
Gravity + Accurate Collisional physics



A shearing box is characterized by its physical parameters

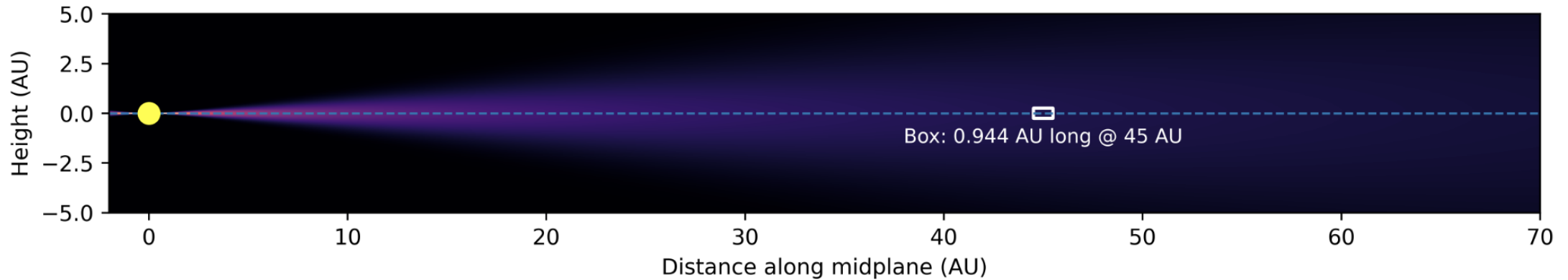
Parameter	Expression	Description
Dimensionless stopping time	$\tau = t_{stop}\Omega$	Relates the particle's aerodynamic and orbital timescale, analog for particle's size.
Dust-to-gas ratio	$Z = \frac{\Sigma_p}{\Sigma_g}$	Ratio of solid surface density to solid gas density.
Radial pressure gradient	$\Pi = \frac{\eta v_K}{c_s}$	The strength of the gas velocity or headwind the particles feel.
Relative strength of self-gravity	$\tilde{G} = \frac{4\pi G \rho_g}{\Omega^2}$	The strength of gravity to be able to hold the disk together vs being sheared apart. Sets the gas mass.

Assumptions: Disk model sets mass scale for simulation



- Planetesimal mass scales by local disk gas density and temperature.
- These values are uncertain for the solar PPD, so we treat disk model as a free parameter
 - Align the simulation sizes with the CCKBOs' size and report corresponding disk parameters that make this work.

Assumptions: Disk model sets mass scale for simulation



$$\Sigma_g = \Sigma_c \left(\frac{r}{R_c} \right)^{-\gamma} \exp \left[- \left(\frac{r}{R_c} \right)^{2-\gamma} \right]$$

$$T = T_{\text{mid},0} \left(\frac{r}{150 \text{ AU}} \right)^{-q} \rightarrow H \text{ depends on } c_s \text{ which depends on } T$$

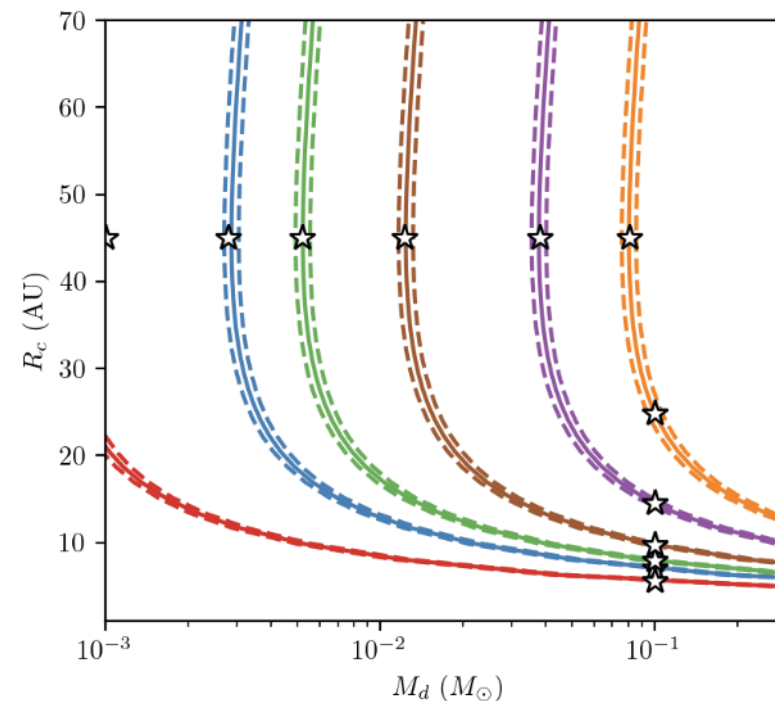
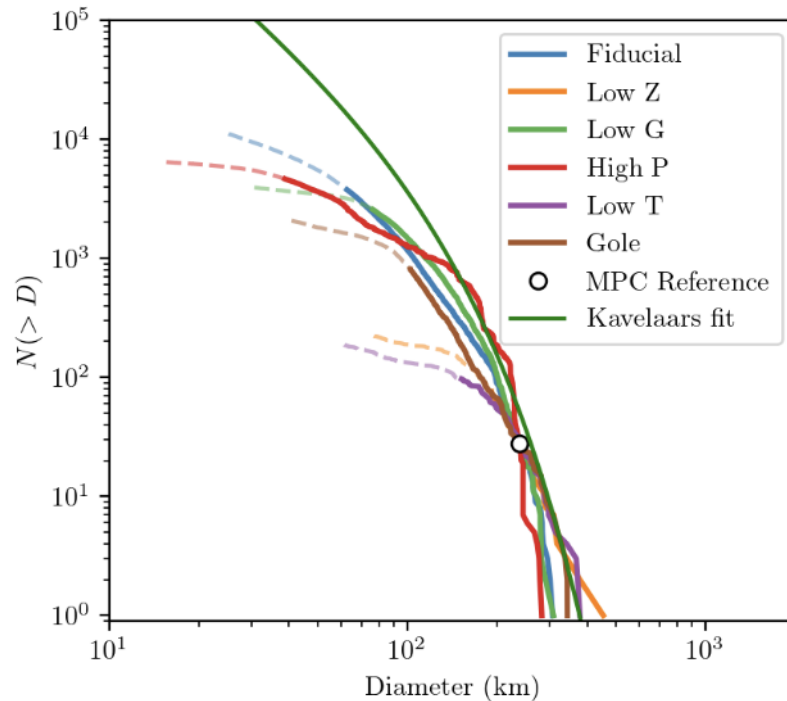
$$\rho_g = \frac{\Sigma_g}{\sqrt{2\pi} H},$$

$$M_H = \rho_g H^3$$

Code mass $\rightarrow$ physical mass

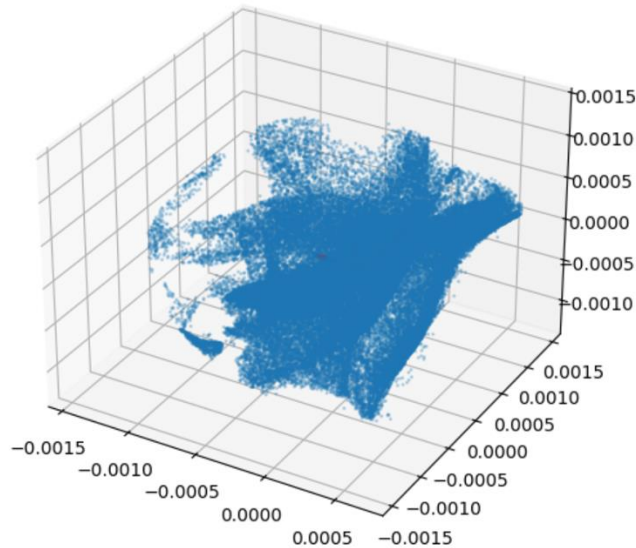
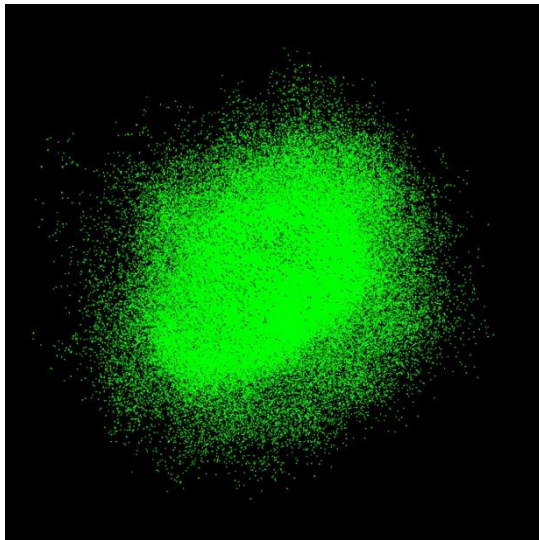
Disk model cont.

- IMF values depend on disk model parameters, that are not well constrained
- In general, the IMFs are going to be aligned with observation, then the disk model parameters calculated and checked for feasibility



Assumptions: Gravitational collapse sets number/size of objects from a clump

- Particle self-gravity is discretized to the gas grid, so particles will not collapse below this scale, making “pebble clouds”
- **Assume: one clump from *Athena* → one planetesimal**
 - Monolithic collapse



Material density ρ

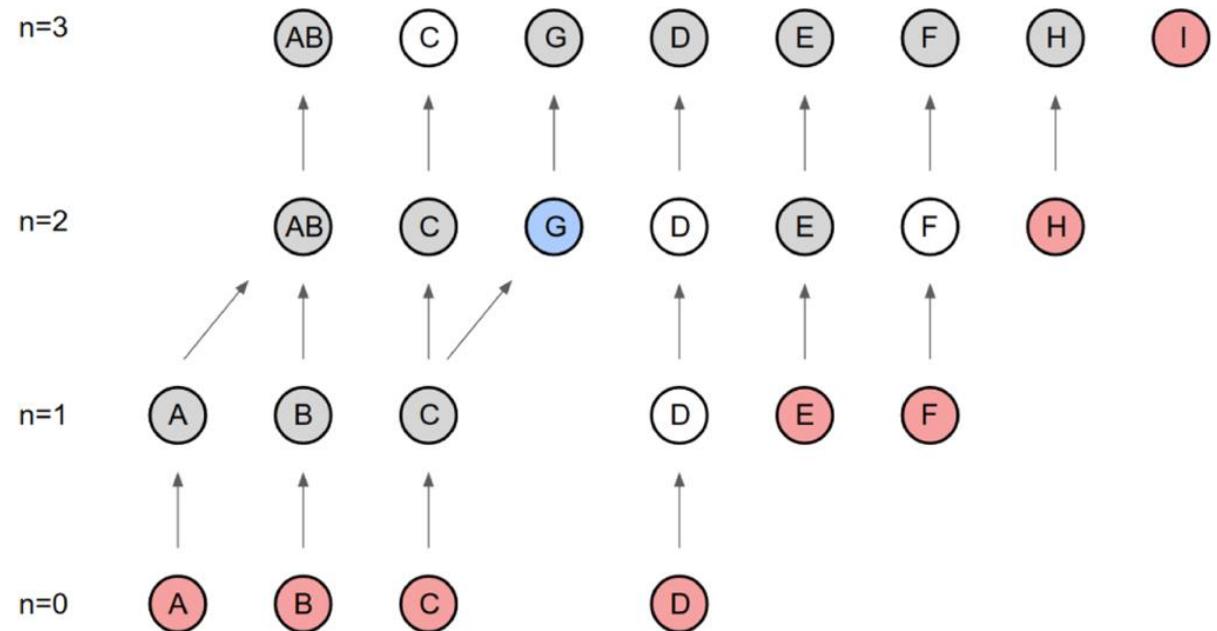
Key improvement: Clump Tracking algorithm

- Previous studies used all masses in one “snapshot” from later in the simulation
 - Misses accretion, mergers, splits, tidal stripping

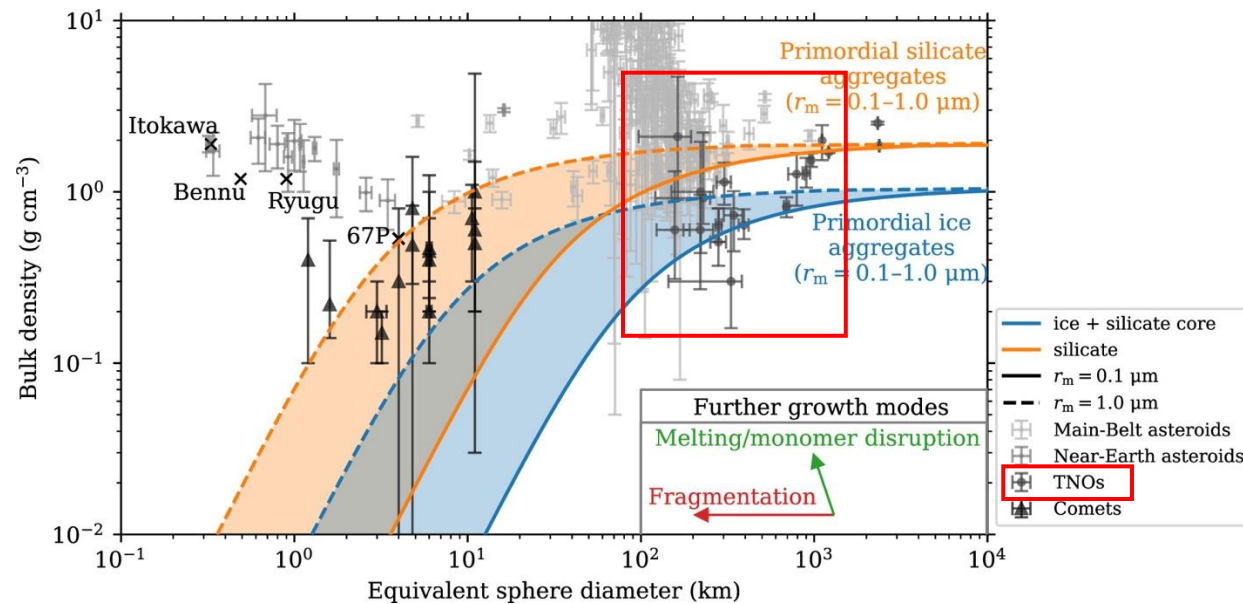
Each circle is a clump identified at time n

The Clump Tracking algorithm assigns labels to particles in the clumps and follows particles from one time to the next

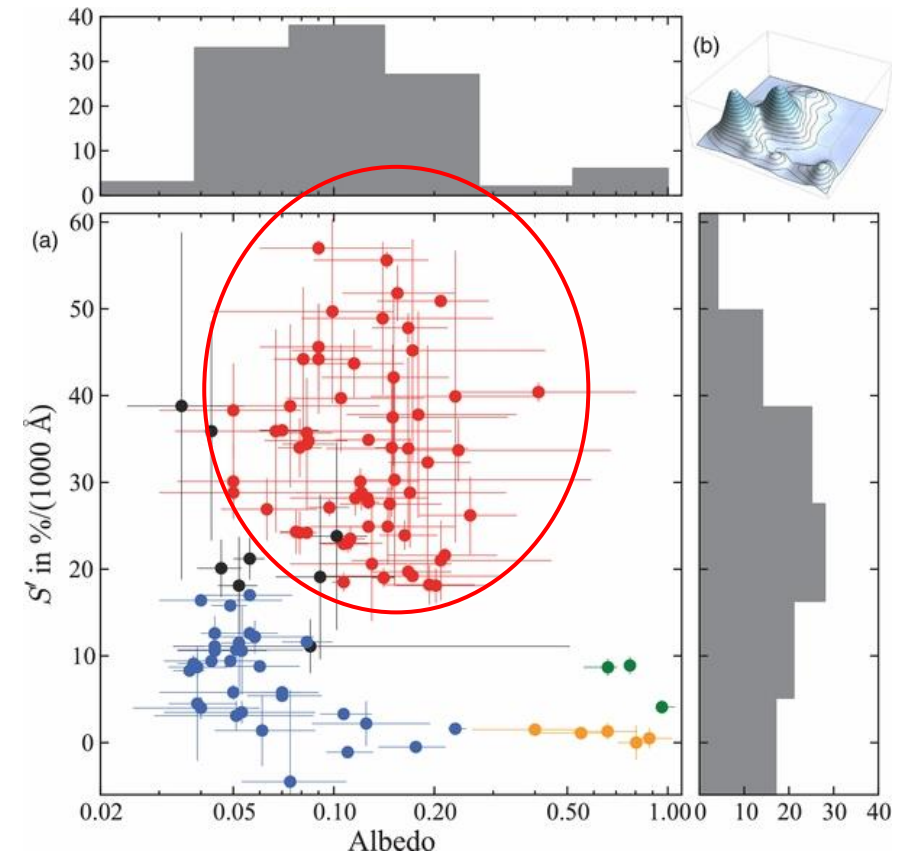
Only clumps in red are “initial” and only they are added to an IMF



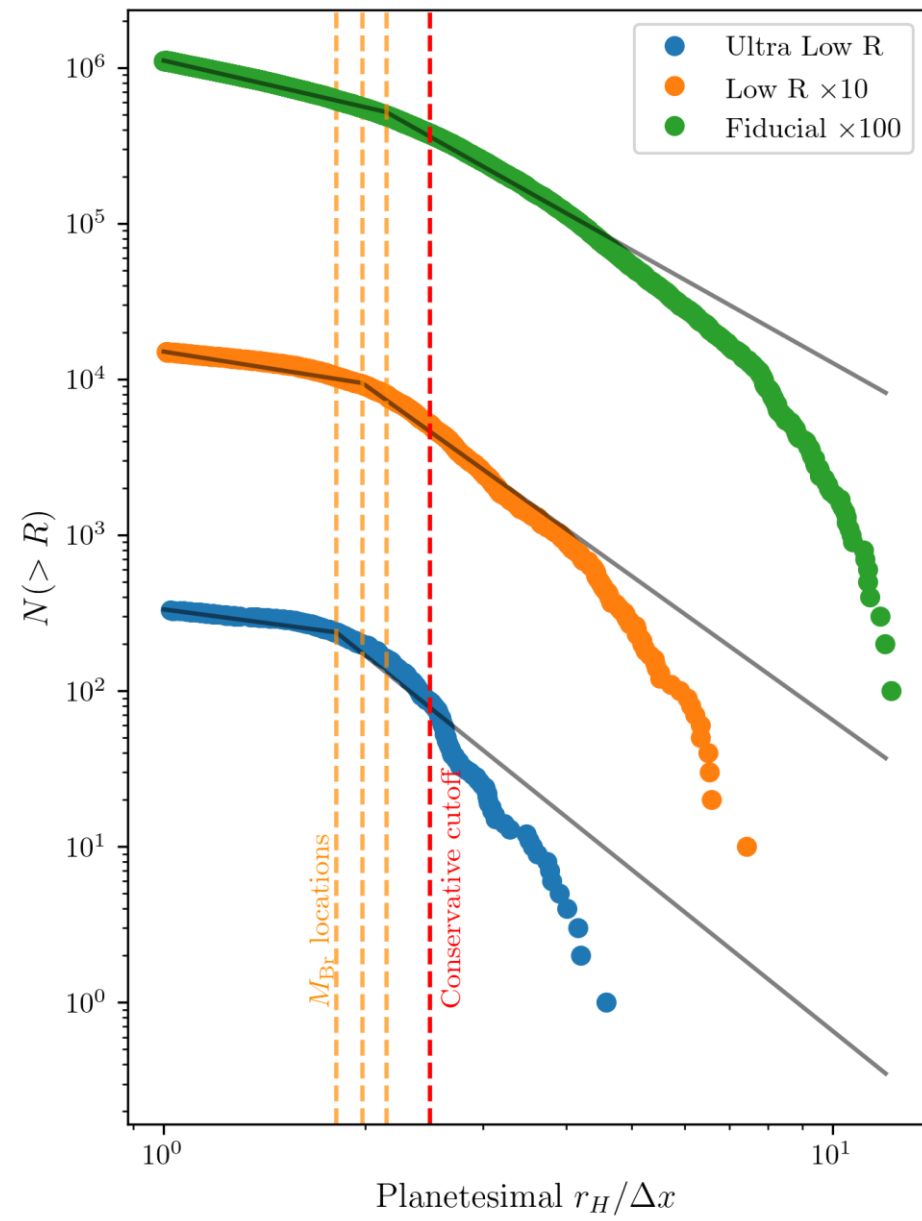
Assumptions: Observational values set albedo for magnitudes and density for mass

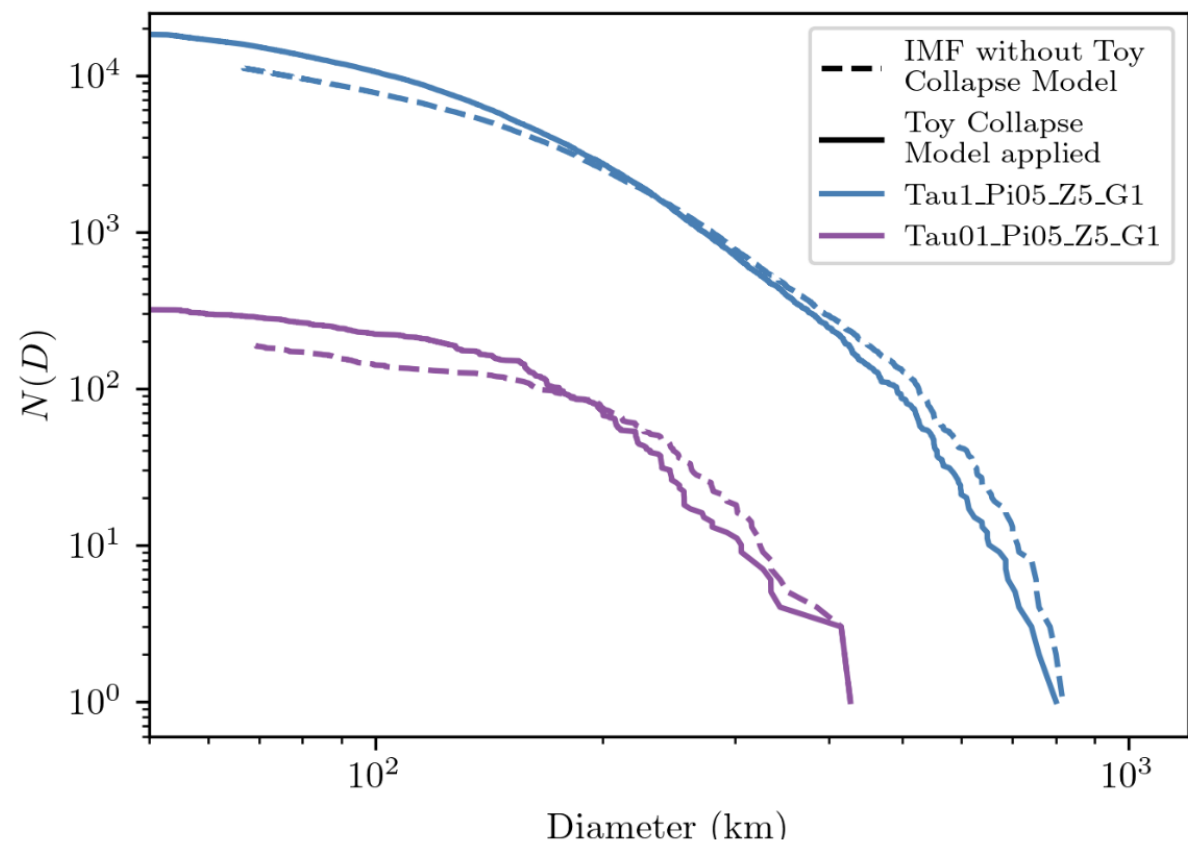


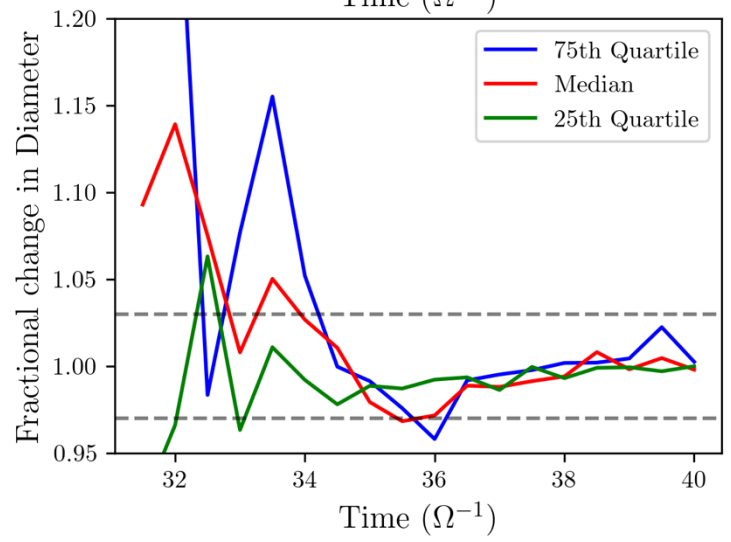
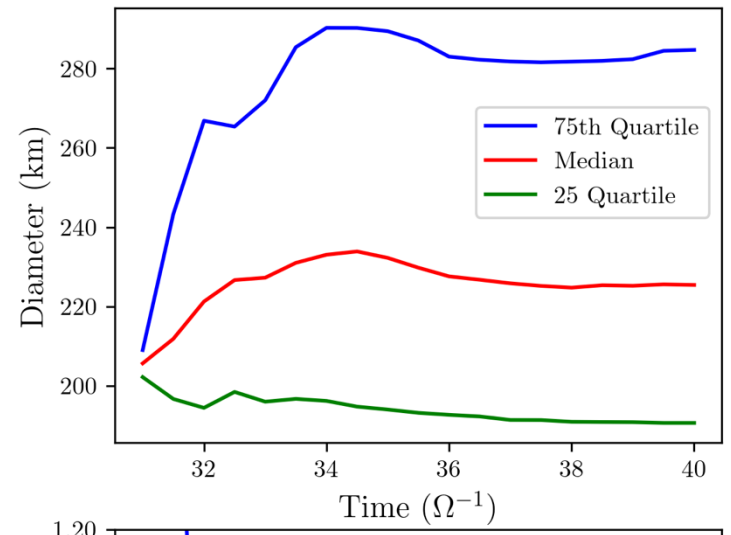
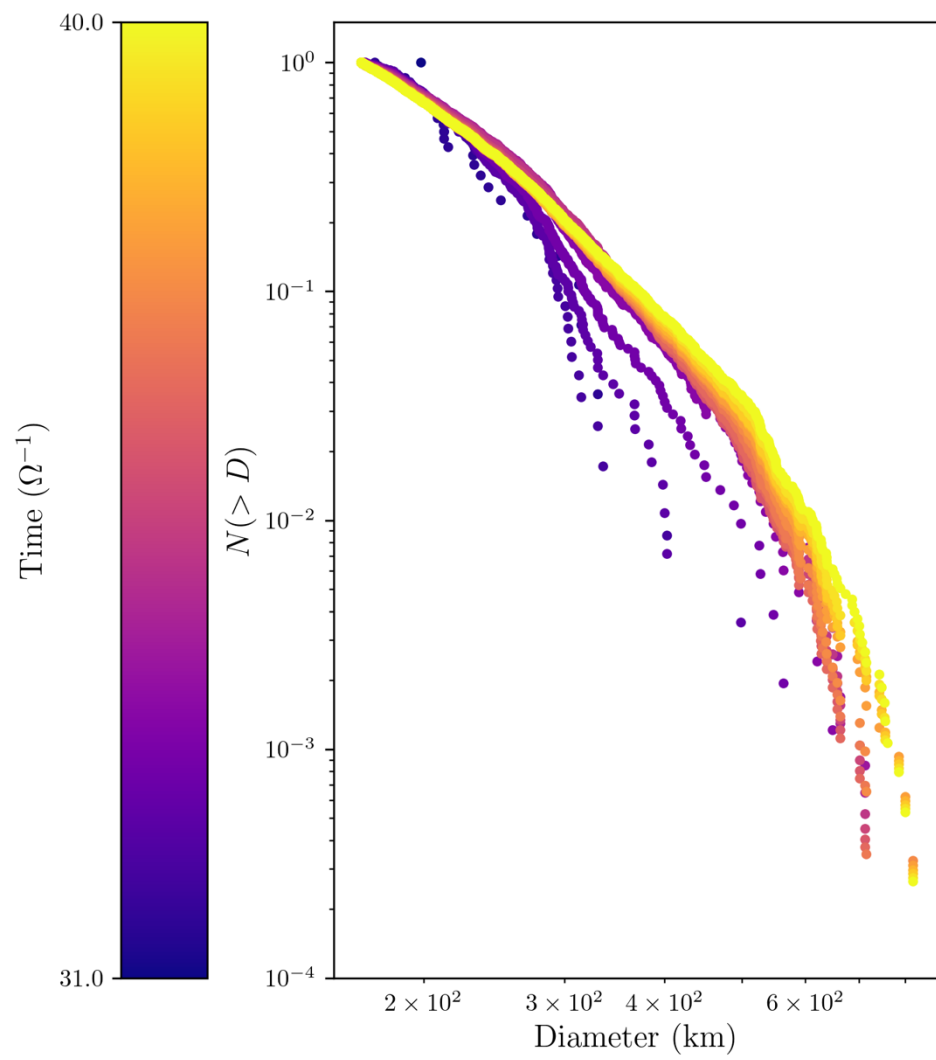
Density $\approx 1 \text{ g/cm}^3$
Up to 120% uncertainty in size

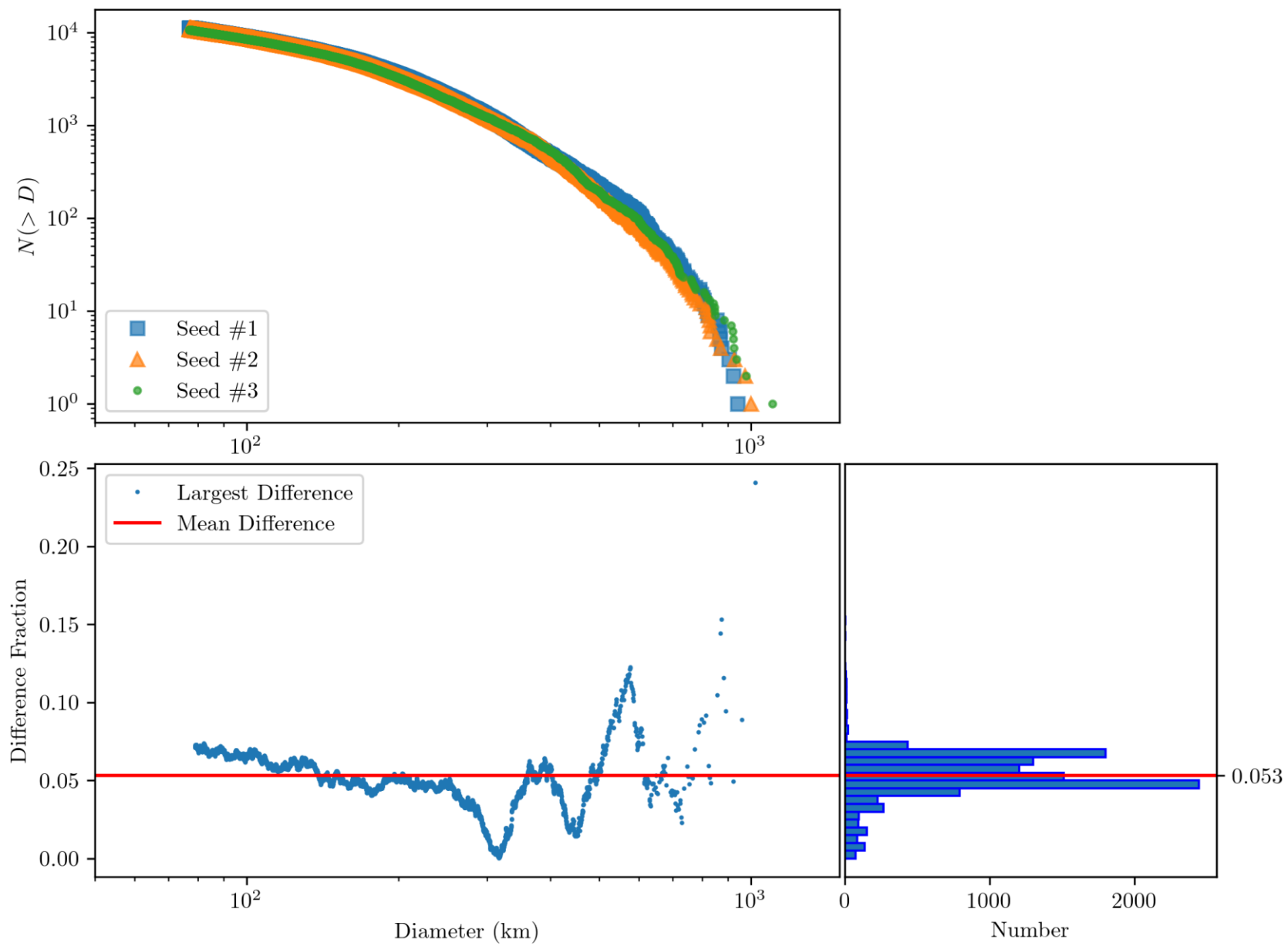


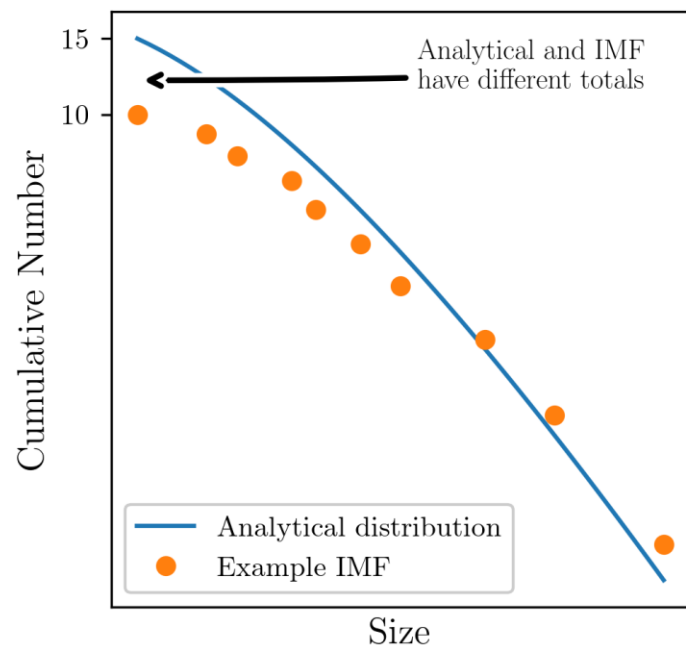
Albedo ≈ 0.15
20% uncertainty in size



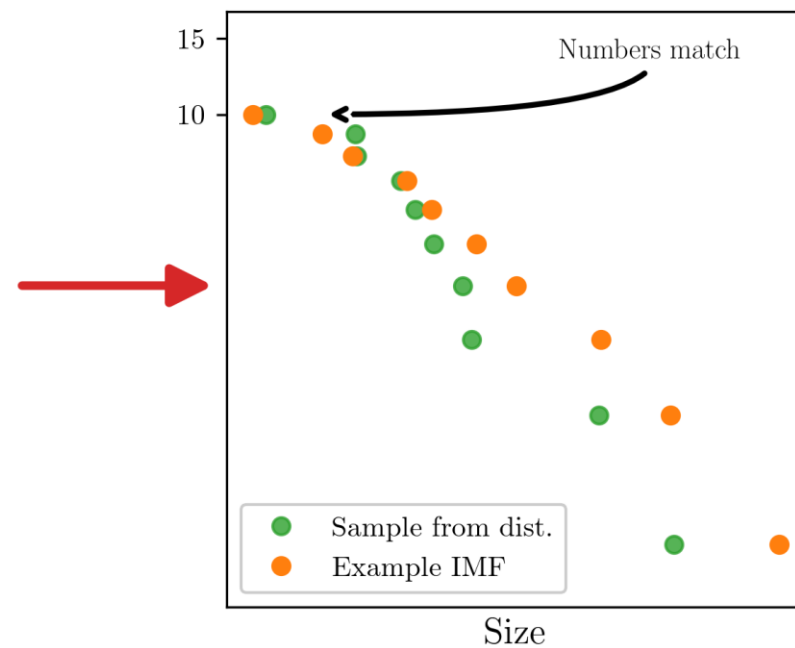
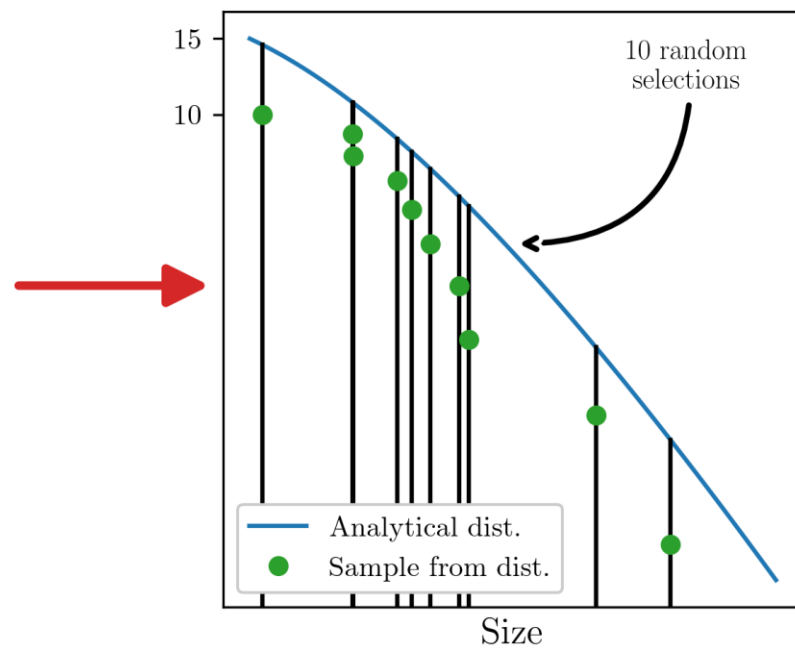








(parametrized comparison)



(sampled comparison)